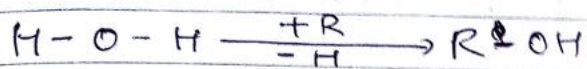


Chapter-3

ALCOHOL (ROH)



→ Alkyl derivative of water is alcohol.

(*) Nomenclature of alcohol.

Common system :- alkyl alcohol.

IUPAC :- alkanol.

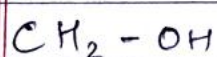
Eg. CH_3OH :- C.S ⇒ Methyl alcohol (wood alcohol)
I.S ⇒ Methanol.

CH_3-CH_2-OH :- C.S ⇒ Ethyl alcohol
I.S ⇒ Ethanol (grain alcohol)

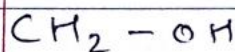
$CH_3-CH_2-CH_2-OH$:- C.S ⇒ n-propyl alcohol.
I.S ⇒ 1-propanol

$\begin{array}{c} CH_3 \\ | \\ CH_3-CH-CH_2-CH_2-OH \end{array}$:- C.S ⇒ iso-pentyl alcohol
I.S ⇒ 3-methyl-1-butanol

$\begin{array}{c} CH_3 \\ | \\ CH_3-C-CH_2-OH \\ | \\ CH_3 \end{array}$:- C.S ⇒ neo-pentyl alcohol.
I.S ⇒ 2,2-dimethyl-1-propanol.

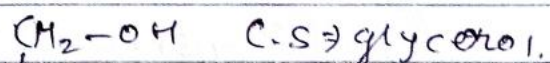


1



C.S ⇒ ethylene glycol.

I.S ⇒ 1,2-ethanediol.



C.S ⇒ glycerol.

I.S ⇒ 1,2,3-propanetriol.

(*) Classification of alcohol.

a) Based on type of carbon to which hydroxyl group is present.

(i) primary (1°) alcohol.
→ OH group bonded with 1° carbon.
eg. $\text{CH}_3 - \text{CH}_2 - \text{OH}$.

(ii) secondary (2°) alcohol.
→ OH group bonded with 2° carbon.
eg. $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$

(iii) Tertiary (3°) alcohol.
→ OH group bonded with 3° carbon.
eg. $\text{CH}_3 - \underset{\text{OH}}{\overset{\text{CH}_3}{\text{C}}} - \text{CH}_3$

b) Based on no. of OH group present.

(i) Monohydric alcohol.
→ only one OH alcohol.
eg. $\text{CH}_3 - \text{CH}_2 - \text{OH}$.

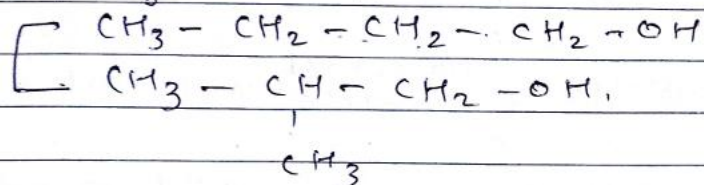
(ii) Dihydric alcohol.
→ two OH group
eg. $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_2 - \text{OH}$

(iii) polyhydric alcohol.
more than 2
→ ~~three~~ OH group.
eg. $\text{CH}_2 - \text{OH}$
|
 $\text{CH} - \text{OH}$
|
 $\text{CH}_2 - \text{OH}$.

ISOMERISM OF ALCOHOL

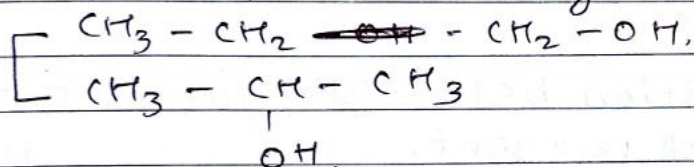
(i) ~~CH₃~~ chain isomerism.

→ Same molecular formula but different chain length.



(ii) position isomerism.

→ Same molecular formula but different position of functional group.

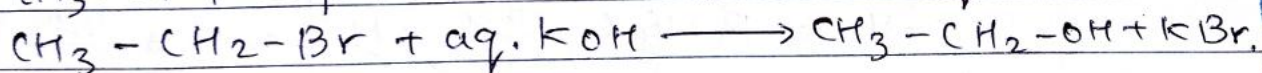
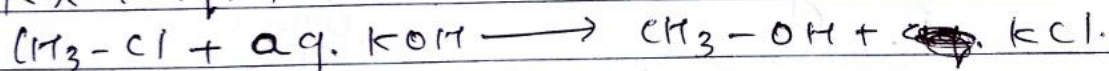
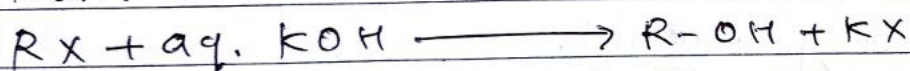


(iii) Functional isomerism.

→ Same M.F but different functional group.

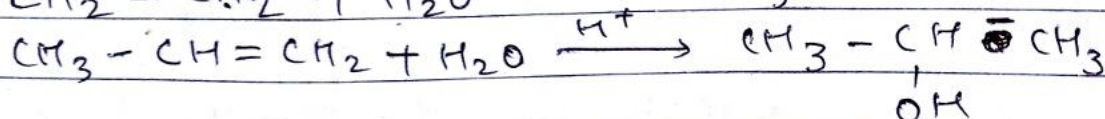
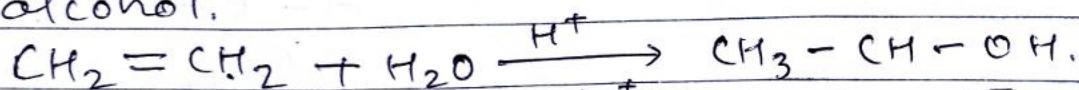
Method of preparation.

(i) From haloalkane.



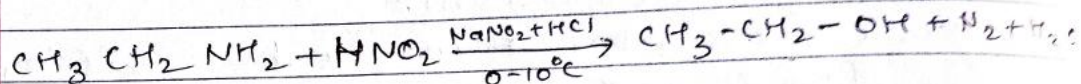
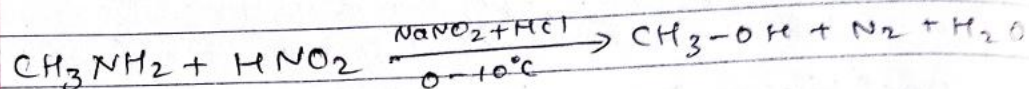
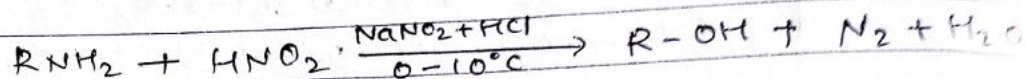
(ii) From alkene.

→ In presence of acid, a molecule of water is added to a molecule of alkene to give alcohol.



(iii) From primary amine.

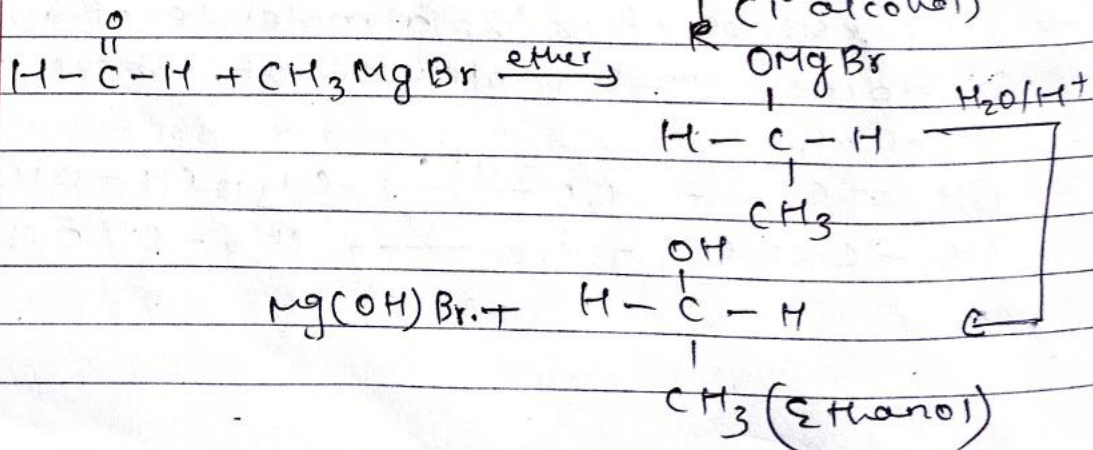
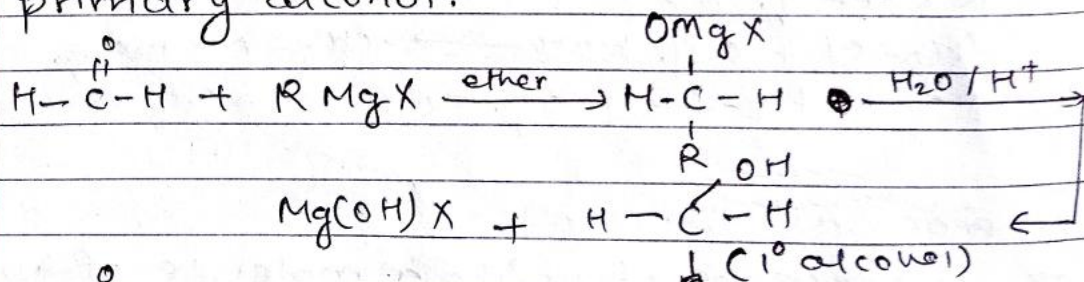
→ Primary amine when treated with nitrous acid under cold condition, alcohol is formed.



(iv) By reaction betⁿ aldehyde or ketone with Grignard reagent.

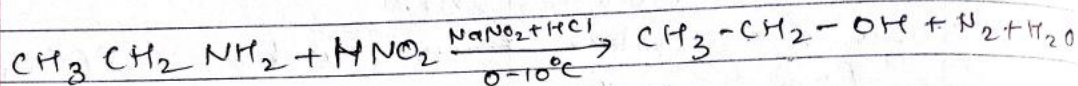
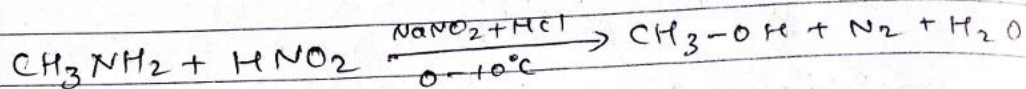
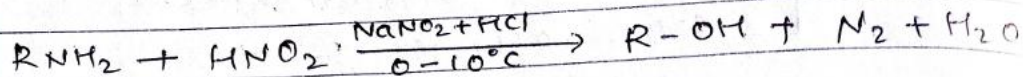
→ When aldehyde or ketone is treated with grignard reagent in presence of ether, an addition product is formed. Which on hydrolysis in acidic medium gives alcohol.

a) Under same condition, formaldehyde gives primary alcohol.



(iii) From primary amine.

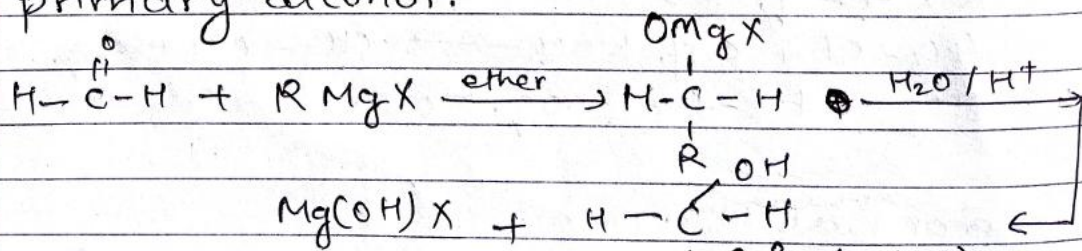
→ primary amine when treated with nitrous acid under cold condition, alcohol is formed.



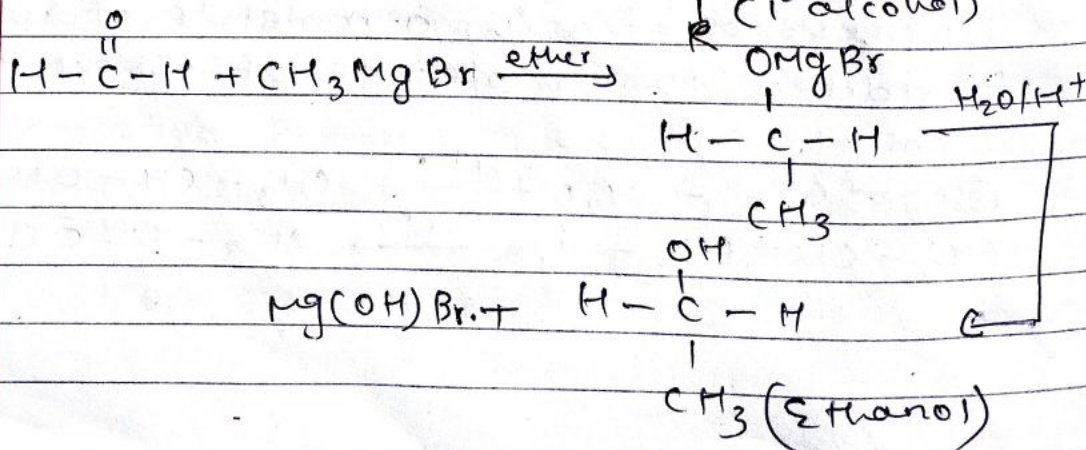
(iv) By reaction betⁿ aldehyde or ketone with Grignard reagent.

→ When aldehyde or ketone is treated with grignard reagent in presence of ether, an addition product is formed. Which on hydrolysis in acidic medium gives alcohol.

a) Under same condition, formaldehyde gives primary alcohol.



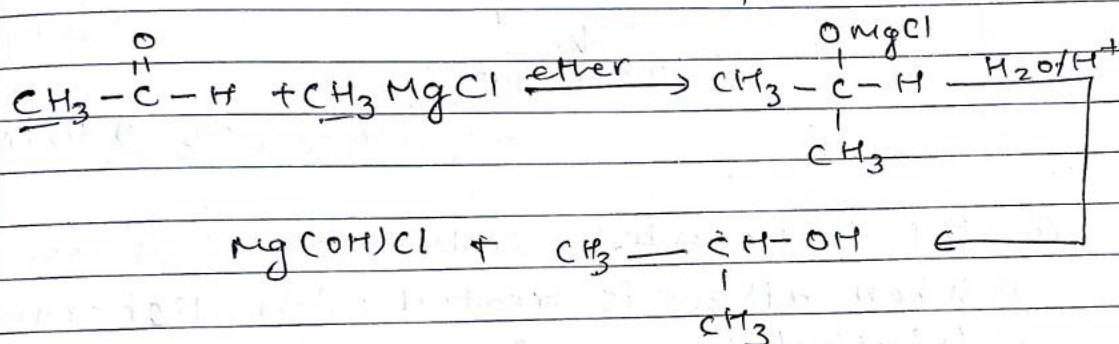
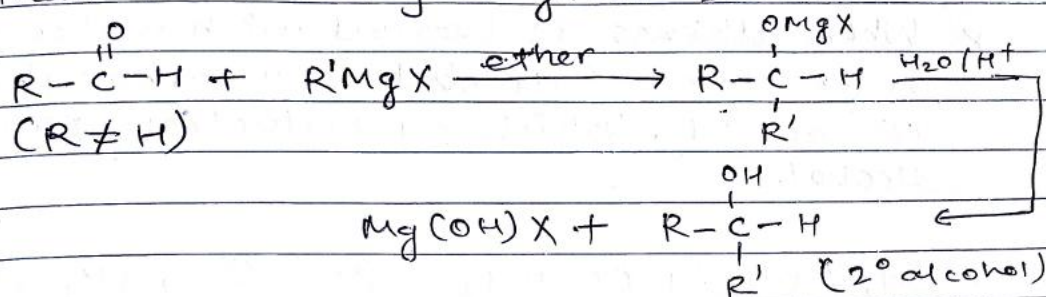
$Mg(OH)X + H-C(H)(OH)R$ (1° alcohol)



$Mg(OH)Br +$

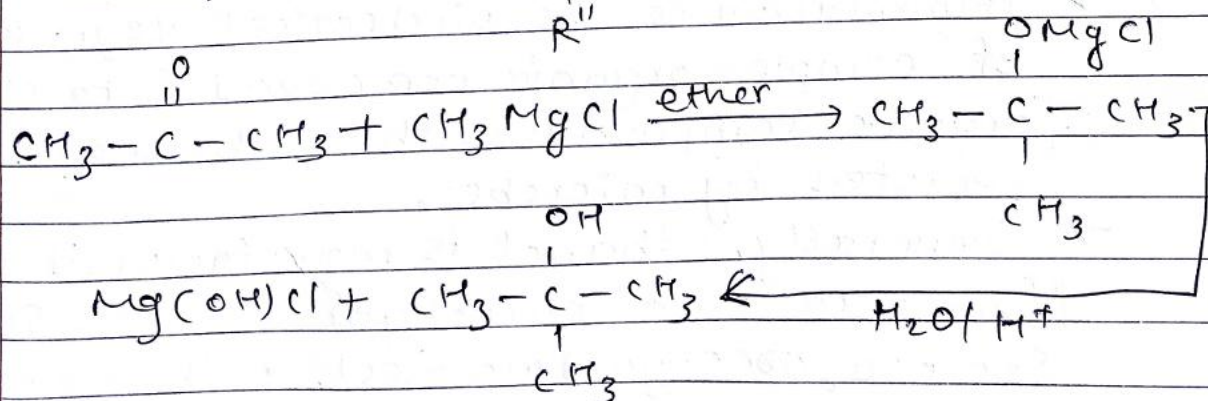
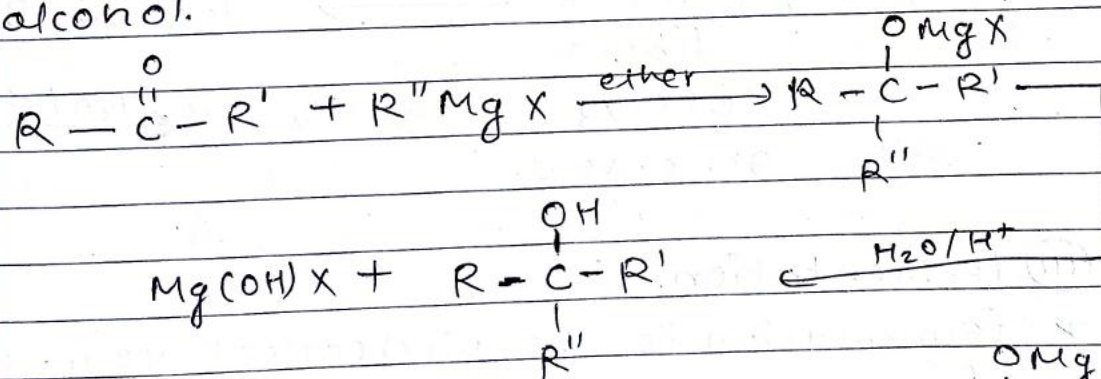
CH_3 (Ethanol)

- b) ~~Under~~ Under same condition, aldehydes other than formaldehyde gives 2° alcohol.



2-propanol

- c) Under same condition, ketone gives tertiary alcohol. (3°)

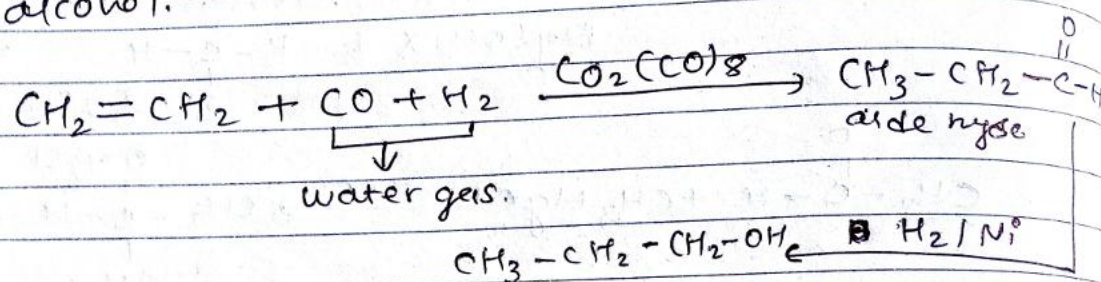


Imp

Manufacture of alcohol.

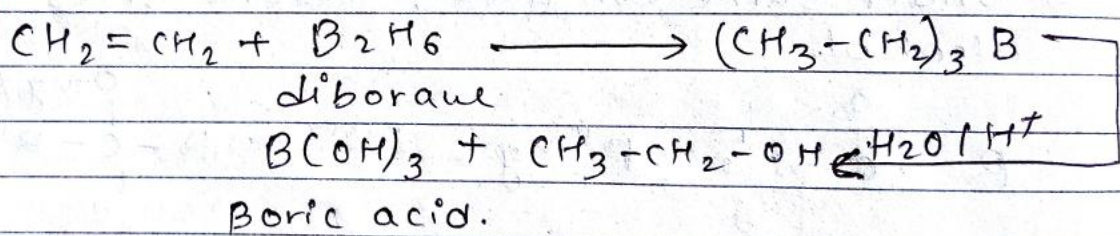
(i) oxo process

→ When alkenes is heated with water gas in presence of dicobalt octa carbonyl (aldehyde) is formed. Which on reduction gives alcohol.



(ii) By ~~hydrocarb~~ hydroboration of alkene.

→ When alkene is treated with diborane, trialkyl borane is obtained, which on oxidation by Hydrogen peroxide in alkaline medium gives alcohol.

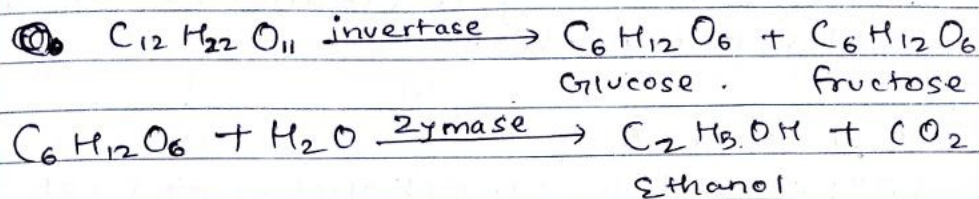


(iii) Fermentation.

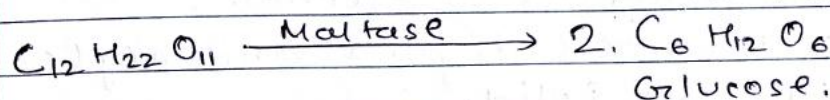
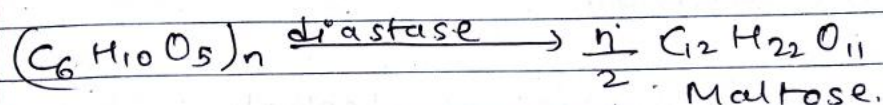
→ Fermentation is the biochemical degradation of complex organic compound into simple organic compound with the help of enzymes, secreted by microbes.

→ Generally, ethanol is manufactured by fermentation of carbohydrates like Starch, Sucrose, (sugar/molasses), with the help of enzymes secreted by yeast.

a) Fermentation of sucrose.



b) Fermentation of starch.



✓ Different forms of alcohol (ethanol)

① Absolute alcohol.

→ Alcohol having 100% ethanol is called as absolute alcohol.

→ Absolute alcohol is obtained by dehydration of rectified spirit with the help of quick lime (CaO).

② Rectified spirit.

→ Alcohol having 95% ethanol is called as Rectified spirit.

→ Rectified spirit is obtained by distillation of fermented alcohol.

(iii) Denaturated alcohol.

→ Alcohol containing methanol is called as Denaturated alcohol

(iv) Power alcohol.

→ 80% petrol + 20% ethanol → fuel.

(v) Alcoholic Beverage.

→ Edible alcohol.

Physical Properties of alcohol.

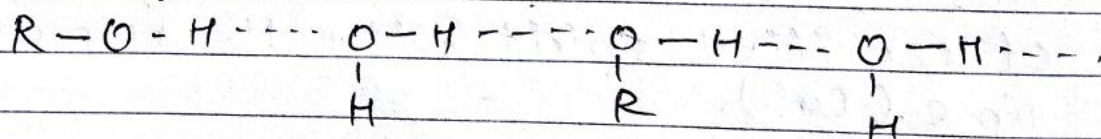
→ Lower members liquid and higher members are solid.

→ Colourless.

→ Characteristics alcoholic smell.

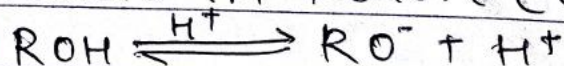
→ M.P. and B.P higher than corresponding haloalkanes because alcohols can forms hydrogen bond and extra energy is required to break that bond.

→ Lower members are highly soluble in water because alcohol can forms hydrogen bond with H₂O molecules.

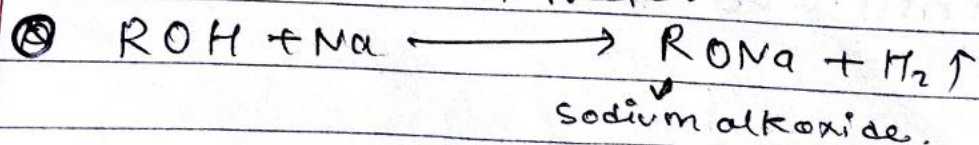


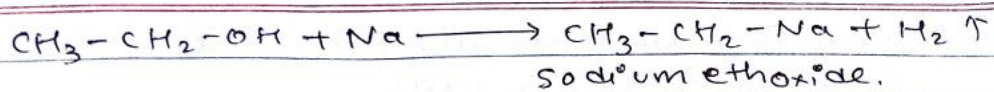
Chemical properties of alcohol.

(i) Acidic in Nature (weak acid).

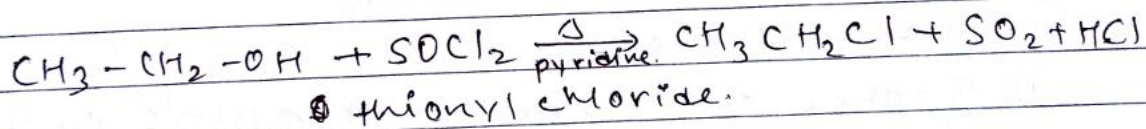
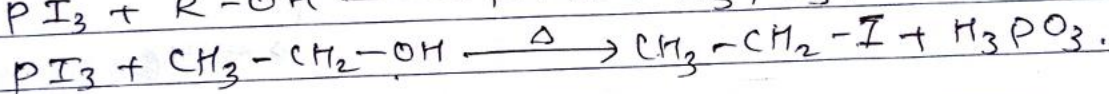
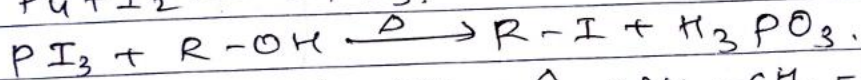
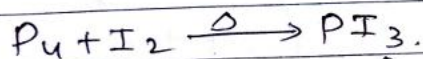
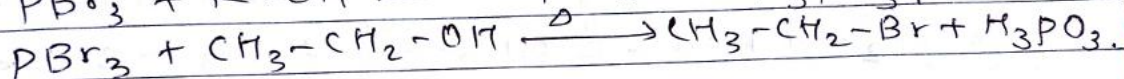
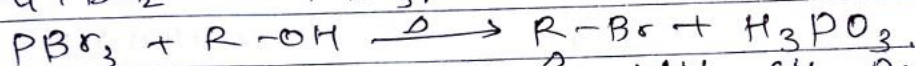
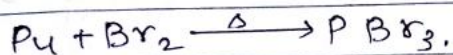
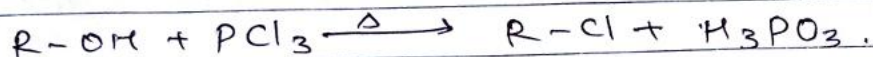
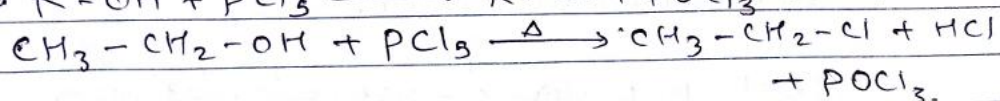
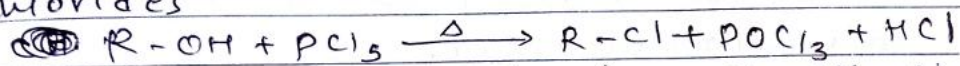


(ii) Rxn with sodium metal.

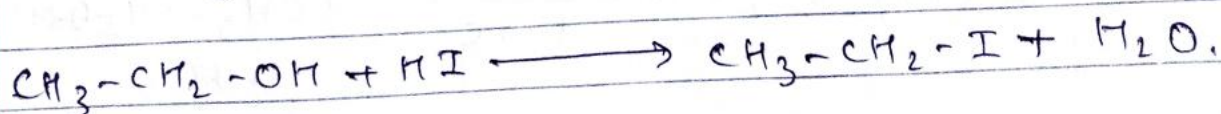
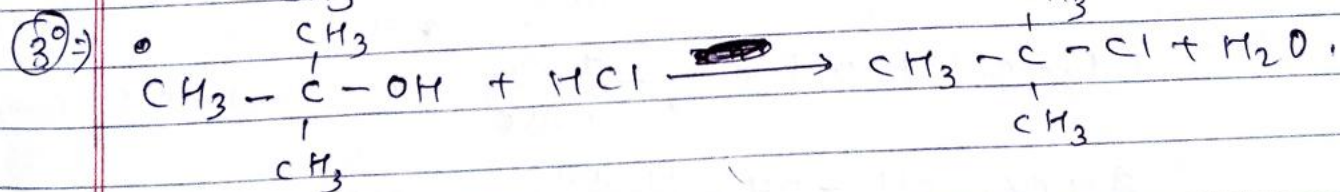
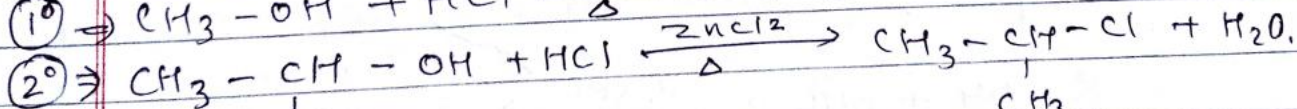
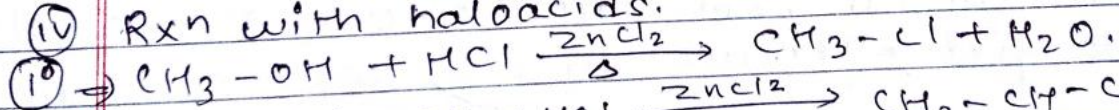




(ii) Rxn with phosphorus halides and thionyl chlorides



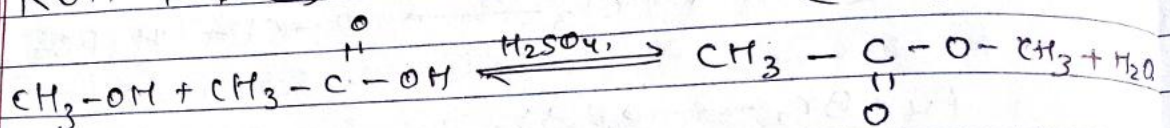
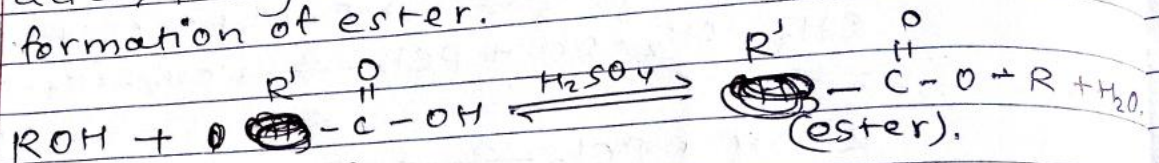
(iv) Rxn with haloacids.



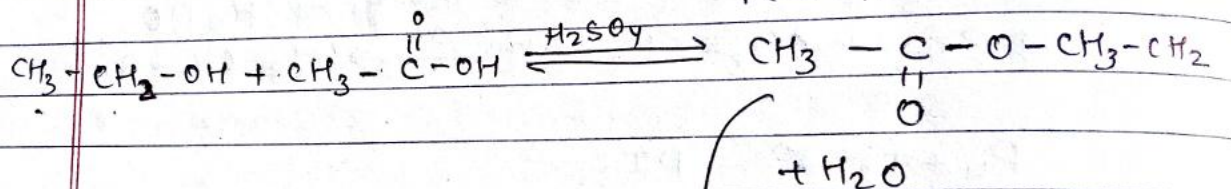
(V) Reaction with carboxylic acid.

(Esterification rxn).

→ When alcohol is treated with carboxylic acid in presence of conc. Sulphuric acid, fruity smell is obtained due to formation of ester.



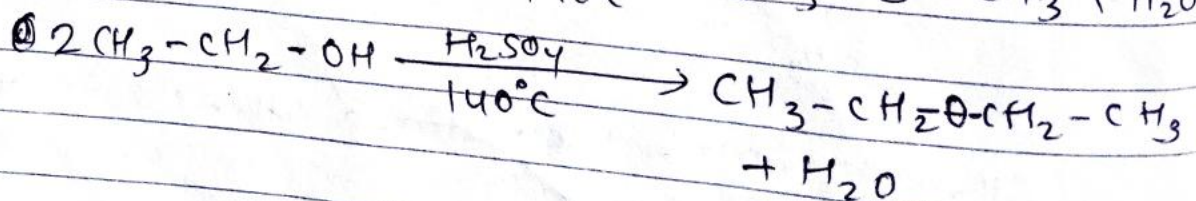
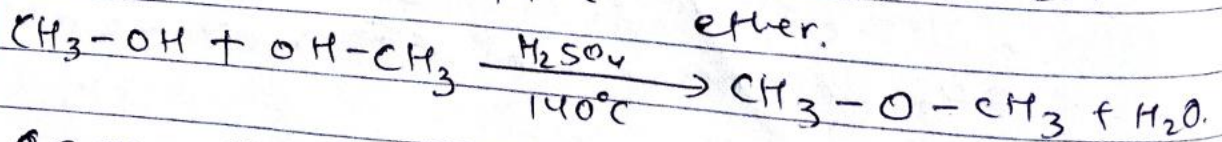
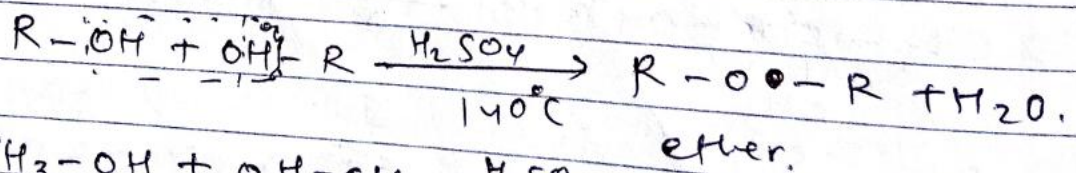
Methyl methanoate.



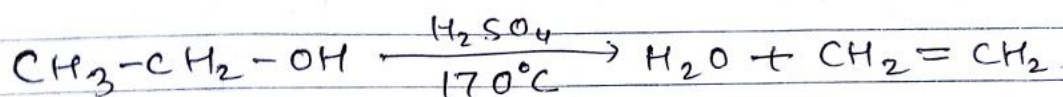
+ H₂O
→ ~~Methoxy ethane~~ Methoxy ethane

(VI) Action of H₂SO₄.

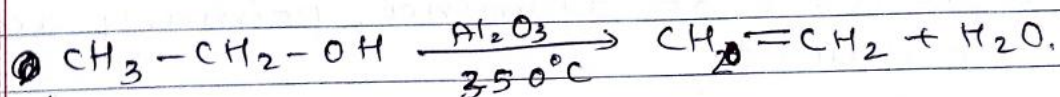
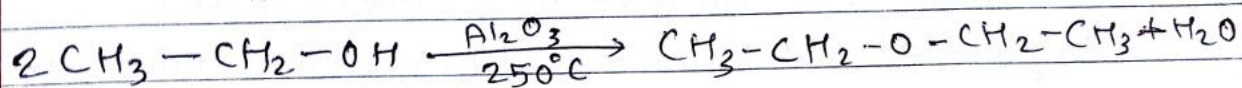
a) Alcohol when heated up to 140°C in presence of conc. Sulphuric acid, a molecule of water is removed from two molecules of alcohol to give ether.



- b) Alcohol on heating with H_2SO_4 upto 170°C , a molecule of water is removed from a molecule of alcohol to give alkene.

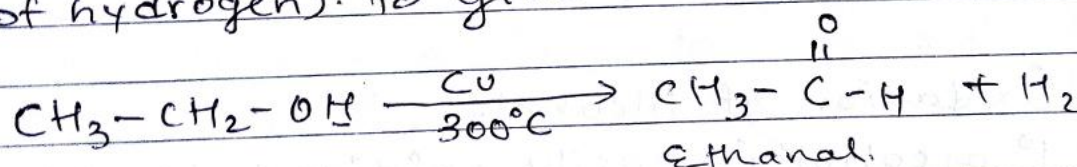


(VII) Action of Alumina (Al_2O_3).

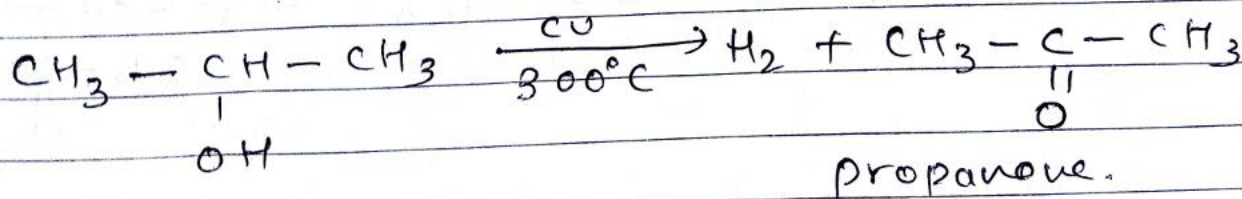


(VIII) Action of copper.

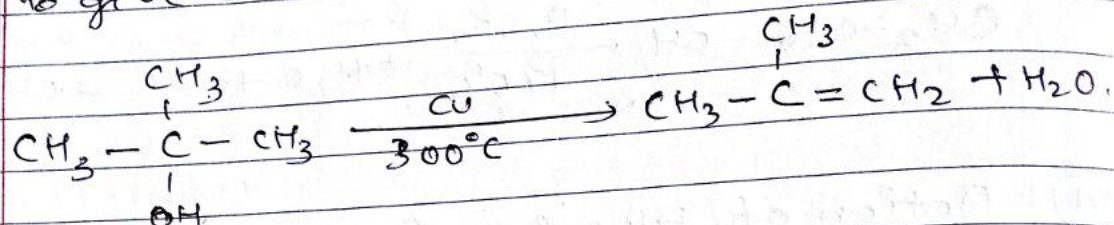
- a) primary alcohol on heating with copper upto 300°C undergoes dehydrogenation (loss of hydrogen) to give aldehyde.



- b) Secondary alcohol on heating with copper upto 300°C undergoes dehydrogenation to give ketones.

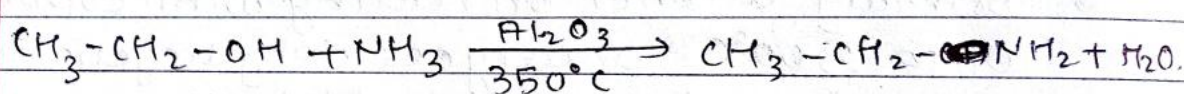
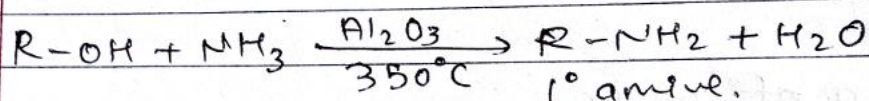


c) Tertiary alcohol when heated upto 300°C in presence of copper undergoes dehydration to give alkenes.



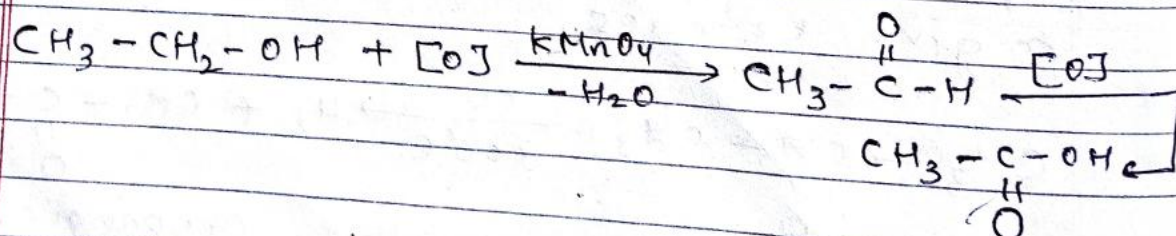
(ix) Rxn with ammonia. (Ammunolysis)

→ Alcohol when strongly heated with ammonia in presence of alumina, primary amine is obtained.



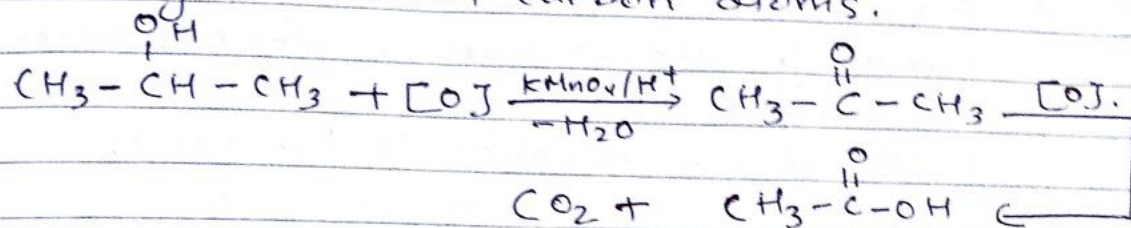
(x) Oxidation of Alcohol.

→ 1° alcohol is easily oxidized to aldehyde by simple oxidizing agent. Aldehyde on further oxidation gives carboxylic acid having same no. of carbon atoms.

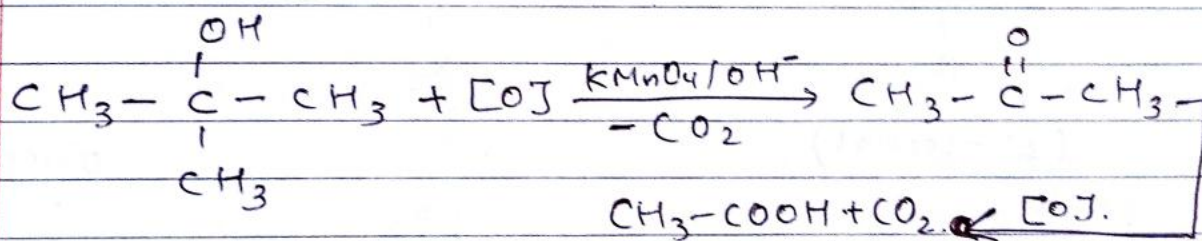


→ 2° alcohol is oxidized to corresponding ketones by strong oxidizing agent which on further oxidation gives carboxylic acid

having less no. of carbon atoms.



→ Tertiary alcohol is resistant towards oxidation reaction but under vigorous condition, it is oxidized to give ketone having less no. of carbon atoms. Ketone on further oxidation gives carboxylic acid having reduced no. of carbon atoms.



Victor Mayer's method to distinguish 1°, 2° and 3° alcohol.

- Victor Mayer's method is the most versatile technique to distinguish primary, secondary and tertiary alcohol.
- In this method, first of all given sample of alcohol is treated with mixture of red phosphorous and iodine to convert into iodo alkane. Then it is treated with silver nitrite and followed by rxn with nitrous acid. Finally aqueous KOH is added to make alkaline.

→ If red colour is obtained, given alcohol is primary. Blue colour indicates secondary alcohol and there is no colouration at all if given alcohol is tertiary.

